

Chapter 2 Test

September 6, 2020

Problem 1

A *bit string* is a list of 0s and 1s. How many length- k bit strings can be made?

Problem 2

A U.S. Social Security number is a nine-digit number. The first digit(s) maybe 0. How many have exactly one 8?

Problem 3

There are six different French books, eight different Russian books, and five different Spanish books. In how many different ways can these books be arranged on a bookshelf if all books in the same language are grouped together?

Problem 4

Either prove the statement is true or give a counterexample to show that it is false. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Problem 5

Either prove the statement is true or give a counterexample to show that it is false. If $A \in B$ and $B \subseteq C$, then $A \subseteq C$.

Problem 6

True or False: Please label each of the following sentences as either *true* or *false*. (You do not need to prove your assertions.)

- a) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, x + y = 0$.
- b) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, x + y = 0$.
- c) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, xy = 0$.
- d) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy = 0$.
- e) $\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, x \leq y$.
- f) $\exists x \in \mathbb{N}, \forall y \in \mathbb{N}, x \leq y$.

Problem 7

Suppose A and B are finite sets. Given that $|A| = 13$, $|A \cup B| = 26$, and $|A \cap B| = 3$. Determine $|B|$.

Problem 8

Let n be a positive integer. Use algebra to simplify the following expression:

$$(x - 1)(1 + x + x^2 + \dots + x^{n-1}).$$

Use this to prove:

$$2^0 + 2^1 + 2^2 + \dots + 2^{n-1} = 2^n - 1.$$

Extra Credit:

Let n be a positive integer. Give a combinatorial proof of the identity

$$n^3 = n(n-1)(n-2) + 3n(n-1) + n.$$